Matrix inequalities and norms

The Lih–Wang permanent inequality toward the flat matrix

Difficulty: extreme **Importance:** interesting to specialist

Provenance: explicit conjecture

Problem statement

For every integer $n \geq 3$, let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ be doubly stochastic: $a_{ij} \geq 0$ and every row and every column has sum one. Put $J_n = \mathbf{1}\mathbf{1}^T/n$, with $\mathbf{1} \in \mathbb{R}^n$ the all-ones vector. Is

$$\operatorname{per}(tJ_n + (1-t)A) \leq t\operatorname{per}(J_n) + (1-t)\operatorname{per}(A) \qquad \text{for every } t \in [1/2, 1]?$$

Here $\operatorname{per}(C) = \sum_{\sigma\in S_n}\prod_{i=1}^{n} c_{i,\sigma(i)}$, so $\operatorname{per}(J_n) = n!/n^n$. Neither symmetry nor positive semidefiniteness is assumed for A.

Relevance

Mixing with the flat matrix is a basic averaging operation on stochastic matrices. The conjecture asks for a sharp bound on the permanent along a specified part of each such segment.

References

1. K.-W. Lih and E. T. H. Wang, *A convexity inequality on the permanent of doubly stochastic matrices*, Congressus Numerantium 36 (1982), 189–198. Original statement and order-three result; bibliographic attribution and formulation reproduced in reference 2, §1.
2. D. K. Udayan and K. Somasundaram, *Lih Wang's and Dittert's conjectures on permanents*, Special Matrices 12 (2024), 20240006, §1, equation (3), Theorems 2.1–2.2. Published full text; older arXiv v1.

Status check — 2026-09-08

The arXiv record remains its December 2023 v1, but the May 2024 published paper has materially narrower claims than that preprint's abstract. Published Theorem 2.1 covers order four; Theorem 2.2 covers only certain order-six matrices and $t \in [0.7836, 1]$. The general-dimensional assertion survives. Searches used *Lih Wang conjecture 2026 permanent*, *Lih-Wang permanent counterexample proof 2025 2026*, and the exact journal title. No full resolution was located. The full journal theorem statements were read; the 1982 paper was traced through their references. The failed extension to all $t \in [0, 1]$ is a different assertion.